Urban flooding and population exposure analysis in Jakarta

A GIS-Based Assessment of Flood Risk, Drainage Infrastructure Gaps, and Vulnerable Districts

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1. Introduction

- Jakarta faces recurring urban flooding due to dense population, river systems, and low-lying terrain.
- This project uses GIS and spatial analysis techniques to map flood risk across Jakarta districts.
- Multiple spatial factors including flood history, elevation, population, and drainage infrastructure were analyzed.
- The study identifies high-risk and infrastructure-deficient areas

2. Problem Statement

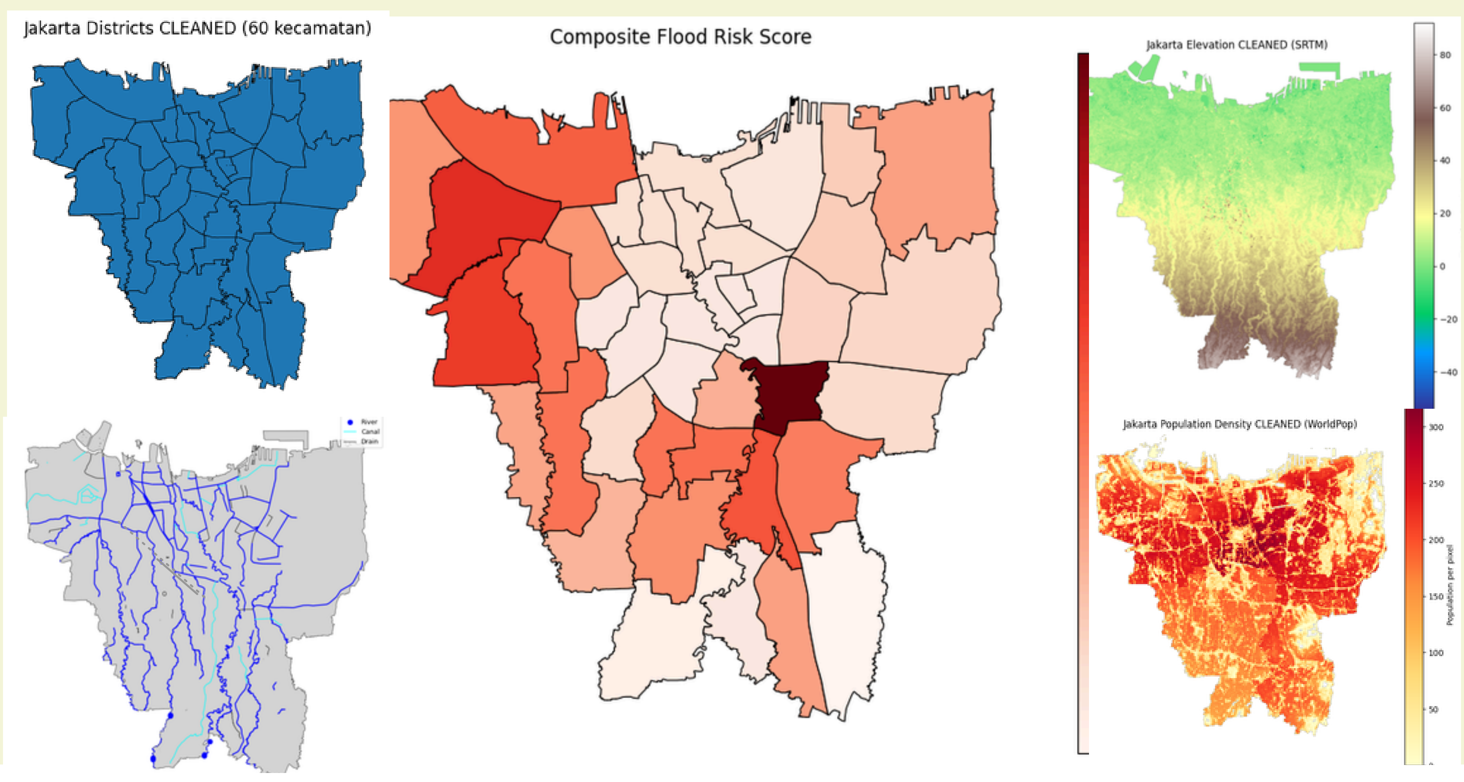
- Which districts in Jakarta are most vulnerable to urban flooding?
- How are flood risk patterns related to population exposure and drainage infrastructure coverage?
- Which high-risk districts should be prioritized for future drainage infrastructure and shelter improvements?

3. Data

- **SRTM Elevation Raster** - NASA EarthExplorer
- **Population Density Raster** - WorldPop
- **Flood Hazard Index 2023** - DKI Jakarta Open Data Portal
- **Flood Incident Records 2016–2020** - DKI Jakarta Open Data Portal
- **District Boundaries** - OpenStreetMap via OSMnx
- **Drainage/River Network** - OpenStreetMap via Overpass API

3.1 Cleaning

- All datasets were reprojected to the common projected coordinate reference system **EPSG:3857** for meter-based distance calculations.
- **Invalid** and **NoData** raster values were masked and removed using null assignment **np.nan** values across data arrays.
- Continuous topographical parameters were transformed into ordinal datamulti-criteria evaluation (MCE) vulnerability scores using jenks, scaled from 0 to 4.
- Localized drainage metrics were computed across administrative geometries into a weighted network density index mapping vector features per **KM squared**



4. Methodology

- **Data Collection** — Five spatial datasets acquired: SRTM elevation raster (NASA EarthExplorer), WorldPop population raster, flood hazard index and incident records 2016–2020 (DKI Jakarta Open Data), and drainage network (OpenStreetMap via OSMnx)
- **Raster Processing** — Elevation tiles merged, rasters clipped to Jakarta mainland boundary, NoData values removed, zonal statistics extracted per kecamatan
- **MCE Scoring** — Each indicator scored 0–4 using Jenks and weighted: 0.45 × Incidents + 0.30 × Hazard + 0.15 × Elevation + 0.10 × River Proximity
- **Spatial Weights Matrix** — Queen contiguity matrix constructed (n=42, avg 5.1 neighbours), row-standardised
- **Spatial Autocorrelation** — Global Moran's I, Local Moran's I (LISA), and Getis-Ord Gi* applied to detect clusters and hotspots
- **Spatial Regression** — OLS baseline with LM diagnostics, followed by ML Spatial Lag model to account for neighbourhood spillover.
  1. **OLS (Elevation coefficient = 1.12, Drainage coefficient = -7.7, R sq. = 0.27)**
  2. **ML lag (Elevation coefficient = 0.31, Drainage coefficient = -4.88, R sq. = 0.43)**
  3. **W_incidents = 0.529 (p = 0.001)**
- **Infrastructure Gap Analysis** — Drainage density computed per km^2, overlaid with risk tiers to identify priority investment zones

5. Results & Discussion

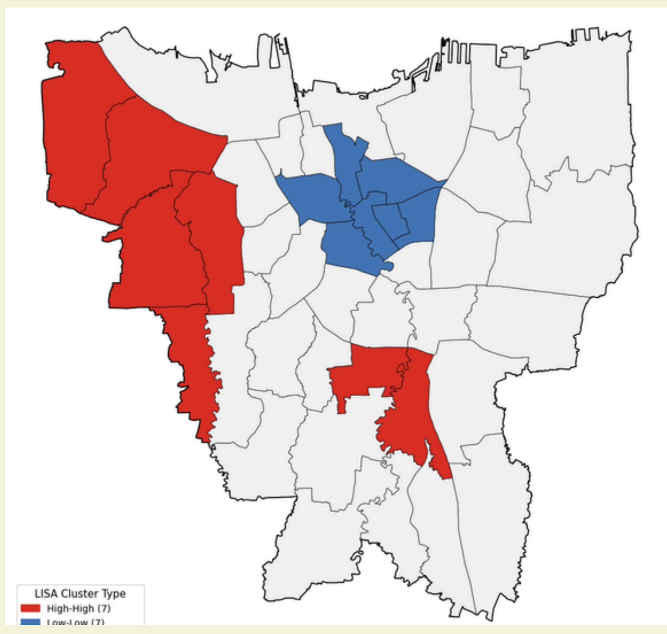


Fig.1 LISA plotting

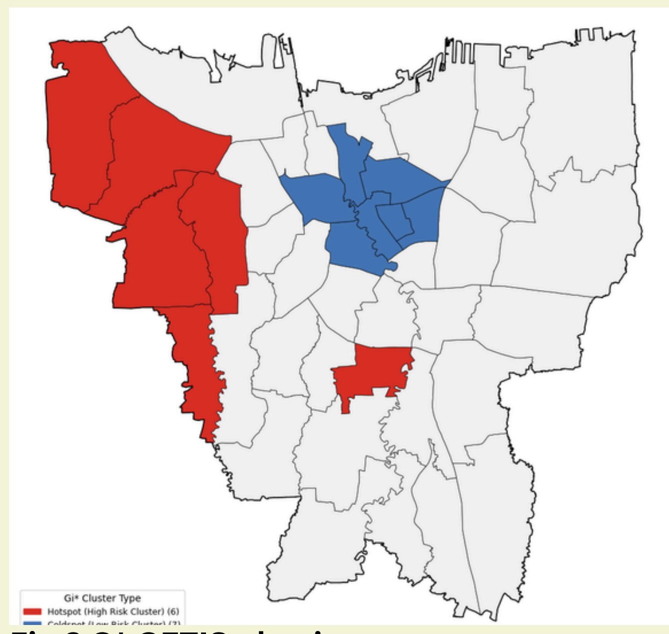


Fig.2 G* GETIS plotting

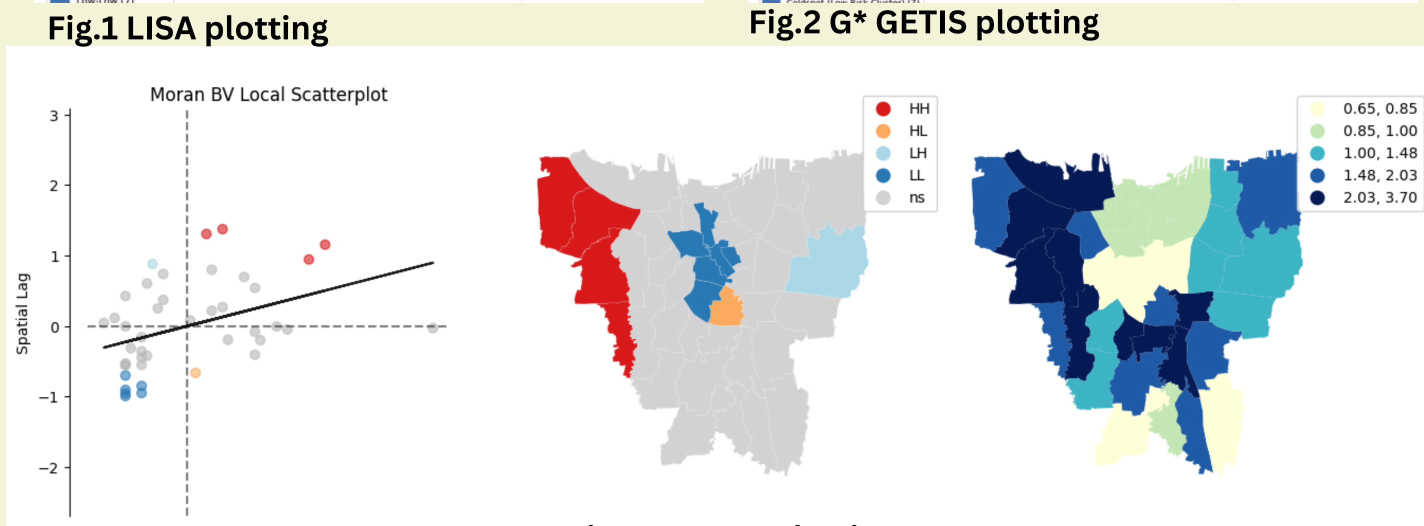
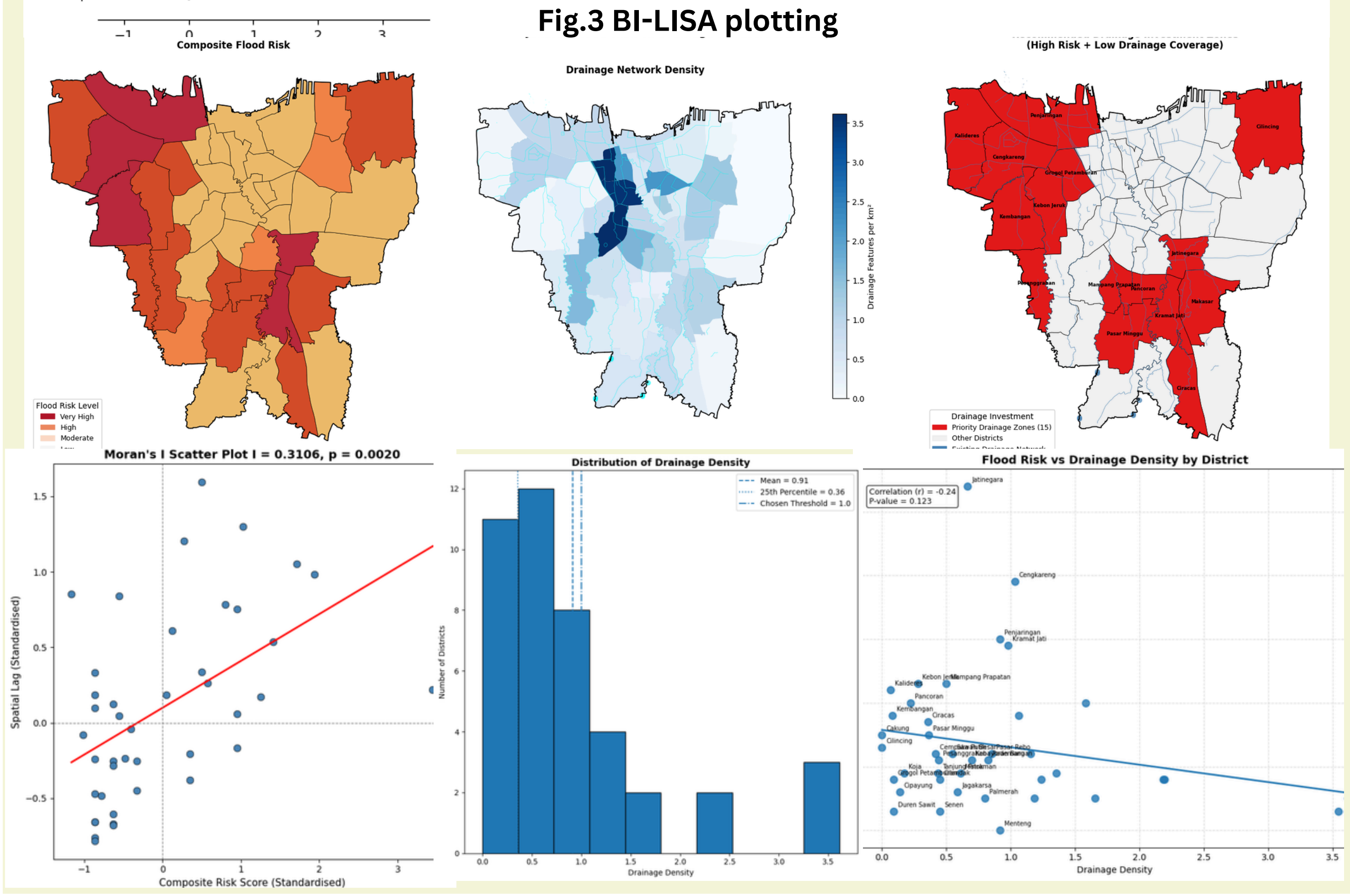


Fig.3 BI-LISA plotting



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